E-COMMERCE SALES DATA ANALYSIS AND DASHBOARD DEVELOPMENT

BY

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GROUP NAME: 3MTT C2 ADVANCED DATA ANALYSIS 102

INTRODUCTION

This project focuses on analyzing an e-commerce dataset to extract meaningful insights that support business decision-making. The dataset initially contained inconsistencies, missing values, and formatting issues, which required thorough data cleaning before analysis.

OBJECTIVES

The main objectives of this project are:

- To transform the e-commerce dataset into a clean, structured format.
- To extract actionable business insights that can improve revenue, customer retention and operational efficiency.
- To analyze sales performance across different dimensions (City, Age, Time).
- To identify key business trends and patterns.
- To build an interactive dashboard for decision-making.

Key Business Questions include:

- Which regions generate the most revenue?
- What product categories perform best?
- What are the peak sales periods?
- How do discounts affect revenue?
- What customer segments drive the most sales?

DATASET DESCRIPTION

The dataset contains transactional data including:

- Order details (order_id, order_date, order_time)
- Customer information (customer_id, customer_age)
- Product details (product_id, category)
- Sales data (quantity, unit_price, discount_pct)
- Location (region, shipping_city)
- Additional fields (rating, return_flag, payment_method)

ecommerce_messy_dataset - Excel (Product Activation Failed)

order_id	customer_id	product_id	order_date	order_time	region	category	quantity	unit_price	discount_pct	payment_method	customer_age	rating	return_flag	shipping_cit
ORD001382	CUST00230	PROD0156	10/17/2022	22:51:03	West	Beauty	9	399.15	7.8	Credit Card	35	3.1	No	DALLAS
ORD001002	CUST00176	PROD0197	24/09/2023	13:43:20	Central	home & gar	8	105.48	16	Bank Transfer	38	1.9	Yes	Phoenix
ORD002682	CUST00138	PROD0103	11/1/2022	8:34:16	East	Electronics	4	587.89	37.5	Credit Card	44	2.5	No	MIAMI
ORD002322		PROD0178	10/2/2023	13:36:06	Central	Home & Gar	5	198.49	28.6	Bank Transfer	72	1.5	No	Phoenix
ORD001669	CUST00188	PROD0178	14/05/2023	19:32:47	South	Beauty	2	-50	33.2	Cash	52	3.6	No	Los Angeles
ORD002153	CUST00220	PROD0144	4/12/2022	19:32:03	West	Beuaty	8	319.79	39.5	Cash	37	3.5	No	Miami
ORD001917	CUST00063	PROD0004	7/22/2023	16:19:57	East	beauty	3	365.82	11.6	Cash		3.2	No	Denver
ORD002116	CUST00354	PROD0076	4/2/2022	18:07:30	East	Beauty	5	47.33	36.3	Bank Transfer	55	2.5	No	Phoenix
ORD002046	CUST00257	PROD0179	8/16/2022	6:51:44	NORTH	Beauty	9	223.51	31	Cash	47	2.7	No	Atlanta
ORD001495	CUST00503	PROD0022	13/01/2022	2:34:05	West	Sports	8	25.08	18.2	Credit Card	24	4.3	No	New York
ORD002304	CUST00073	PROD0137	25/04/2023	21:35:13	north	Toys	5	-50	0.9	Bank Transfer	30	2.2	Yes	Houston
ORD001111	CUST00505	PROD0019	7/12/2023	13:10:28	South	Toys	2	222.95	38.1	Bank Transfer	150	1.1	No	Miami
ORD001654	CUST00181	PROD0185	12/1/2023	13:22:08	south	Electronics	3	753.05	26.4	Cash	73	3.5	No	Houston
ORD001292	CUST00553	PROD0114	1/22/2023	22:20:11	EAST	Sports	4	395.47	2.1	Cash	30	1.3	No	Houston
ORD002402	CUST00271	PROD0082	19/02/2023	16:19:57	South	Books	1	$357.63	0.2	Cash	70	3.2	No	Chicago
ORD001742	CUST00387	PROD0141	11/7/2022	16:13:46	East	Electronics	2	241.07	14.3	Debit Card	26	2.6	No	Seattle
ORD001937	CUST00568	PROD0195	11/7/2023	18:29:22	East	Sports	9	150.79	3.9	Credit Card	72	1.9	No	houston
ORD001711	CUST00430	PROD0068	1/6/2023	11:02:59	North	Spoorts	7	222.53	35.8	PayPal	32	4.4	No	chicago
ORD002504	CUST00032	PROD0181	3/9/2023	11:58:13	South	Books	1	45.79	1.2	PayPal	50	2	No	Denver
ORD001158	CUST00261	PROD0143	8/27/2022	17:23:35	North	Books	8	255.65	14	PayPal	-5	2	No	Denver
ORD002317	CUST00389	PROD0057	3/10/2023	15:20:30	south	Sports	1	118.21	2.2	Cash	40	1.6	No	Denver
ORD002353	CUST00067	PROD0072	2/23/2022	7:05:39	South	Beauty	8	228.51	16	PayPal	26	3	No	Atlanta
ORD002592	CUST00177	PROD0128	6/4/2023	15:22:20	North	TOYS	9	USD 326.25	32.4	Debit Card	29	1.4	No	NEW YORK
ORD001972	CUST00245	PROD0171	8/13/2022	20:29:55	West	Beauty	2	USD 227.86	12.4	Cash	32	1.2	Yes	Phoenix
ORD002546	CUST00420	PROD0104	10/3/2022	6:39:16	south	Books	10	277.09	18.1	PayPal	49	4.2	No	CHICAGO
ORD002008	CUST00462	PROD0099	1/17/2023	7:51:57	east	Beauty	1	150.85	21.1	PayPal	47	2.2	No	miami
ORD001217	CUST00555	PROD0006	3/16/2023	15:12:04	East	Sports	8	145	13.4	Credit Card	59	1.9	No	Phoenix

localhost:8888/notebooks/Documents/E_Commerce.ipynb

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1820 rows × 15 columns

[14]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1820 entries, 0 to 1819
Data columns (total 15 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   order_id        1820 non-null   object
 1   customer_id     1760 non-null   object
 2   product_id      1820 non-null   object
 3   order_date      1820 non-null   object
 4   order_time      1820 non-null   object
 5   region          1753 non-null   object
 6   category        1764 non-null   object
 7   quantity        1820 non-null   int64
 8   unit_price      1820 non-null   object
 9   discount_pct    1740 non-null   float64
 10  payment_method  1761 non-null   object
 11  customer_age    1746 non-null   float64
 12  rating          1716 non-null   float64
 13  return_flag     1820 non-null   object
 14  shipping_city   1820 non-null   object
dtypes: float64(3), int64(1), object(11)
memory usage: 213.4+ KB
```

Filling Missing Values

DATA CLEANING PROCESS

This was the most critical phase of the project.

Handling Missing Values

- Filled categorical nulls with “Unknown”
- Imputed numerical values using mean/median

Fixing Data Types

- Converted unit_price from text to numeric
- Combined order_date and order_time into a datetime column

Handling Inconsistent Data

- Standardized text (e.g., city names, categories)
- Corrected invalid age values:
  - -5 → 50
  - 150 → 15
  - 200 → 20

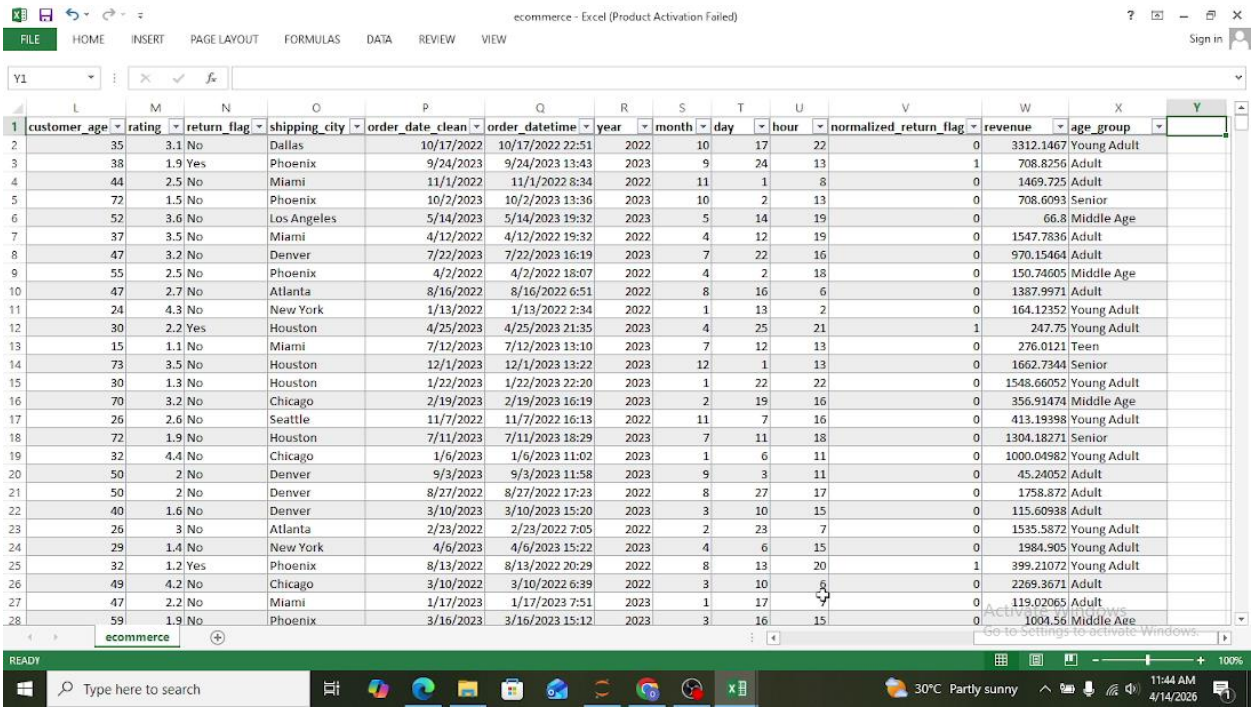
Removing Duplicates

- Identified duplicate order_id values
- Retained the most relevant records

FEATURE ENGINEERING

New features were created to enhance analysis:

- **Revenue Column:**
  Revenue = Quantity × Unit Price × (1 - Discount%)
- **Datetime Features:**
  Year, Month, Day, Hour
- **Customer Segmentation:**
  Age groups (Teen, Young Adult, Adult, Senior, etc.)



customer_age	rating	return_flag	shipping_city	order_date_clean	order_datetime	year	month	day	hour	normalized_return_flag	revenue	age_group
35	3.1	No	Dallas	10/17/2022	10/17/2022 22:51	2022	10	17	22	0	3312.1467	Young Adult
38	1.9	Yes	Phoenix	9/24/2023	9/24/2023 13:43	2023	9	24	13	1	708.8256	Adult
44	2.5	No	Miami	11/1/2022	11/1/2022 8:34	2022	11	1	8	0	1469.725	Adult
72	1.5	No	Phoenix	10/2/2023	10/2/2023 13:36	2023	10	2	13	0	708.6093	Senior
52	3.6	No	Los Angeles	5/14/2023	5/14/2023 19:32	2023	5	14	19	0	66.8	Middle Age
37	3.5	No	Miami	4/12/2022	4/12/2022 19:32	2022	4	12	19	0	1547.7836	Adult
47	3.2	No	Denver	7/22/2023	7/22/2023 16:19	2023	7	22	16	0	970.15464	Adult
55	2.5	No	Phoenix	4/2/2022	4/2/2022 18:07	2022	4	2	18	0	150.74605	Middle Age
47	2.7	No	Atlanta	8/16/2022	8/16/2022 6:51	2022	8	16	6	0	1387.9971	Adult
24	4.3	No	New York	1/13/2022	1/13/2022 2:34	2022	1	13	2	0	164.12352	Young Adult
30	2.2	Yes	Houston	4/25/2023	4/25/2023 21:35	2023	4	25	21	1	247.75	Young Adult
15	1.1	No	Miami	7/12/2023	7/12/2023 13:10	2023	7	12	13	0	276.0121	Teen
73	3.5	No	Houston	12/1/2023	12/1/2023 13:22	2023	12	1	13	0	1662.7344	Senior
30	1.3	No	Houston	1/22/2023	1/22/2023 22:20	2023	1	22	22	0	1548.66052	Young Adult
70	3.2	No	Chicago	2/19/2023	2/19/2023 16:19	2023	2	19	16	0	356.91474	Middle Age
26	2.6	No	Seattle	11/7/2022	11/7/2022 16:13	2022	11	7	16	0	413.19398	Young Adult
72	1.9	No	Houston	7/11/2023	7/11/2023 18:29	2023	7	11	18	0	1304.18271	Senior
32	4.4	No	Chicago	1/6/2023	1/6/2023 11:02	2023	1	6	11	0	1000.04982	Young Adult
50	2	No	Denver	9/3/2023	9/3/2023 11:58	2023	9	3	11	0	45.24052	Adult
50	2	No	Denver	8/27/2022	8/27/2022 17:23	2022	8	27	17	0	1758.872	Adult
40	1.6	No	Denver	3/10/2023	3/10/2023 15:20	2023	3	10	15	0	115.60938	Adult
26	3	No	Atlanta	2/23/2022	2/23/2022 7:05	2022	2	23	7	0	1535.5872	Young Adult
29	1.4	No	New York	4/6/2023	4/6/2023 15:22	2023	4	6	15	0	1984.905	Young Adult
32	1.2	Yes	Phoenix	8/13/2022	8/13/2022 20:29	2022	8	13	20	1	399.21072	Young Adult
49	4.2	No	Chicago	3/10/2022	3/10/2022 6:39	2022	3	10	6	0	2269.3671	Adult
47	2.2	No	Miami	1/17/2023	1/17/2023 7:51	2023	1	17	7	0	119.02065	Adult
59	1.9	No	Phoenix	3/16/2023	3/16/2023 15:12	2023	3	16	15	0	1004.56	Middle Age

M CODE (POWER QUERY)

- Source: = Csv.Document(File.Contents("C:\Users\DELL\Documents\ecommerce.csv"),[Delimiter=",", Columns=24, Encoding=1252, QuoteStyle=QuoteStyle.None])
- Promote Headers: = Table.PromoteHeaders(Source, [PromoteAllScalars=true])
- Changed Type: = Table.TransformColumnTypes(#"Promoted Headers",{{"order_id", type text}, {"customer_id", type text}, {"product_id", type text}, {"order_date", type text}, {"order_time", type time}, {"region", type text}, {"category", type text}, {"quantity", Int64.Type}, {"unit_price", type number}, {"discount_pct", type number}, {"payment_method", type text}, {"customer_age", Int64.Type}, {"rating", type number}, {"return_flag", type text}, {"shipping_city", type text}, {"order_date_clean", type date}, {"order_datetime", type datetime}, {"year", Int64.Type}, {"month", Int64.Type}, {"day", Int64.Type}, {"hour", Int64.Type}, {"normalized_return_flag", Int64.Type}, {"revenue", type number}, {"age_group", type text}})
- Removed Column: = Table.RemoveColumns(#"Changed Type",{"order_date"})
- Inserted Year: = Table.AddColumn(#"Removed Columns", "Year.1", each Date.Year([order_date_clean]), Int64.Type)
- Removed Columns1: = Table.RemoveColumns(#"Inserted Year",{"year"})
- Inserted Month: = Table.AddColumn(#"Removed Columns1", "Month.1", each Date.Month([order_date_clean]), Int64.Type)
- Inserted Start of Month: = Table.AddColumn(#"Inserted Month", "Start of Month", each Date.StartOfMonth([order_date_clean]), type date)
- Inserted Day Name: = Table.AddColumn(#"Inserted Start of Month", "Day Name", each Date.DayOfWeekName([order_date_clean]), type text)
- Removed Columns2: = Table.RemoveColumns(#"Inserted Day Name",{"Month.1", "Start of Month"})
- Inserted Month Name: = Table.AddColumn(#"Removed Columns2", "Month Name", each Date.MonthName([order_date_clean]), type text)

- Reordered Columns: = Table.ReorderColumns(#"Inserted Month Name",{"order_id", "customer_id", "product_id", "order_time", "region", "category", "quantity", "unit_price", "discount_pct", "payment_method", "customer_age", "rating", "return_flag", "shipping_city", "order_date_clean", "order_datetime", "Year.1", "month", "Month Name", "day", "Day Name", "hour", "normalized_return_flag", "revenue", "age_group"})
- Changed Type1: = Table.TransformColumnTypes(#"Reordered Columns",{{"revenue", Currency.Type}, {"unit_price", Currency.Type}})
- Renamed Columns: = Table.RenameColumns(#"Changed Type1",{{"order_id", "Order_ID"}, {"customer_id", "Customer_ID"}, {"product_id", "Product_ID"}, {"order_time", "Order_Time"}, {"region", "Region"}, {"category", "Category"}, {"quantity", "Quantity"}, {"unit_price", "Unit_Price"}, {"discount_pct", "Discount_Pct"}, {"payment_method", "Payment_Method"}, {"customer_age", "Customer_Age"}, {"rating", "Rating"}, {"return_flag", "Return_Flag"}, {"shipping_city", "Shipping_City"}, {"order_date_clean", "Order_Date"}, {"order_datetime", "Order_Datetime"}, {"Year.1", "Year"}, {"month", "Month"}, {"day", "Day"}, {"hour", "Hour"}, {"normalized_return_flag", "Normalized_Return_Flag"}, {"revenue", "Revenue"}, {"age_group", "Age_Group"}})
- Filtered Rows: = Table.SelectRows(#"Renamed Columns", each [Age_Group] <> null and [Age_Group] <> "")
- Replaced Values …. 21: = Table.ReplaceValue(#"Replaced Value…20","Beautty…Tois","Beauty…Toys",Replacer.ReplaceText,{"Category"})
- Filtered Rows1: = Table.SelectRows(#"Replaced Value21", each true)
- Trimmed Text: = Table.TransformColumns(#"Filtered Rows1",{{"Shipping_City", Text.Trim, type text}})
- Filtered Rows2: = Table.SelectRows(#"Trimmed Text", each true)
- Inserted Start of Month1: = Table.AddColumn(#"Filtered Rows2", "Start of Month", each Date.StartOfMonth([Order_Date]), type date)
- Inserted Start of Year: = Table.AddColumn(#"Filtered Rows2", "Start of Month", each Date.StartOfMonth([Order_Date]), type date)
- Inserted Start of Week: = Table.AddColumn(#"Inserted Start of Year", "Start of Week", each Date.StartOfWeek([Order_Date]), type date)
- Removed Columns3: = Table.RemoveColumns(#"Inserted Start of Week",{"Year", "Month", "Month Name", "Day", "Day Name", "Hour"})
- Added Custom: = Table.AddColumn(#"Removed Columns3", "Actual Revenue", each [Quantity] * [Unit_Price])
- Reorderd Columns1: = Table.ReorderColumns(#"Added Custom",{"Order_ID", "Customer_ID", "Product_ID", "Order_Time", "Region", "Category", "Quantity", "Unit_Price", "Discount_Pct", "Actual Revenue", "Revenue", "Payment_Method", "Customer_Age", "Rating", "Return_Flag", "Shipping_City", "Order_Date", "Order_Datetime", "Normalized_Return_Flag", "Age_Group", "Start of Month", "Start of Year", "Start of Week"})
- Changed Type2: = Table.TransformColumnTypes(#"Reordered Columns1",{{"Actual Revenue", Currency.Type}})
- Renamed Columns1: = Table.RenameColumns(#"Changed Type2",{{"Discount_Pct", "Discount_Percentage"}})
- Sorted Rows: = Table.Sort(#"Renamed Columns1",{{"Discount_Percentage", Order.Ascending}})
- Added Conditional Column: = Table.AddColumn(#"Sorted Rows", "Discount Category", each if [Discount_Percentage] <= 10 then "Very Low Discount" else if [Discount_Percentage] <= 20 then "Low Discount" else if [Discount_Percentage] <= 30 then "High Discount" else if [Discount_Percentage] <= 40 then "Very High Discount" else null)
- Reordered Column2: = Table.ReorderColumns(#"Added Conditional Column",{"Order_ID", "Customer_ID", "Product_ID", "Order_Time", "Region", "Category", "Quantity", "Unit_Price", "Discount_Percentage", "Discount Category", "Actual Revenue", "Revenue", "Payment_Method",

"Customer_Age", "Rating", "Return_Flag", "Shipping_City", "Order_Date", "Order_Datetime", "Normalized_Return_Flag", "Age_Group", "Start of Month", "Start of Year", "Start of Week"})

- Changed Type3: = Table.TransformColumnTypes(#"Reordered Columns2",{{"Discount Category", type text}})

DATA MODELLING

 Implemented DAX measures:

- Order Quantity = SUM('E-commerce'[Quantity])
- Return Rate = DIVIDE(SUM('E-commerce'[Normalized_Return_Flag]), COUNT('E-commerce'[Normalized_Return_Flag]))
- Total Acutal Revenue = SUM('E-commerce'[Actual Revenue])
- Total Customers = DISTINCTCOUNT('E-commerce'[Customer_ID])
- Total Orders = DISTINCTCOUNT('E-commerce'[Order_ID])
- Total Products = DISTINCTCOUNT('E-commerce'[Product_ID])
- Total Returns = SUM('E-commerce'[Normalized_Return_Flag])
- Total Revenue = SUM('E-commerce'[Revenue])

EXPLORATORY DATA ANALYSIS

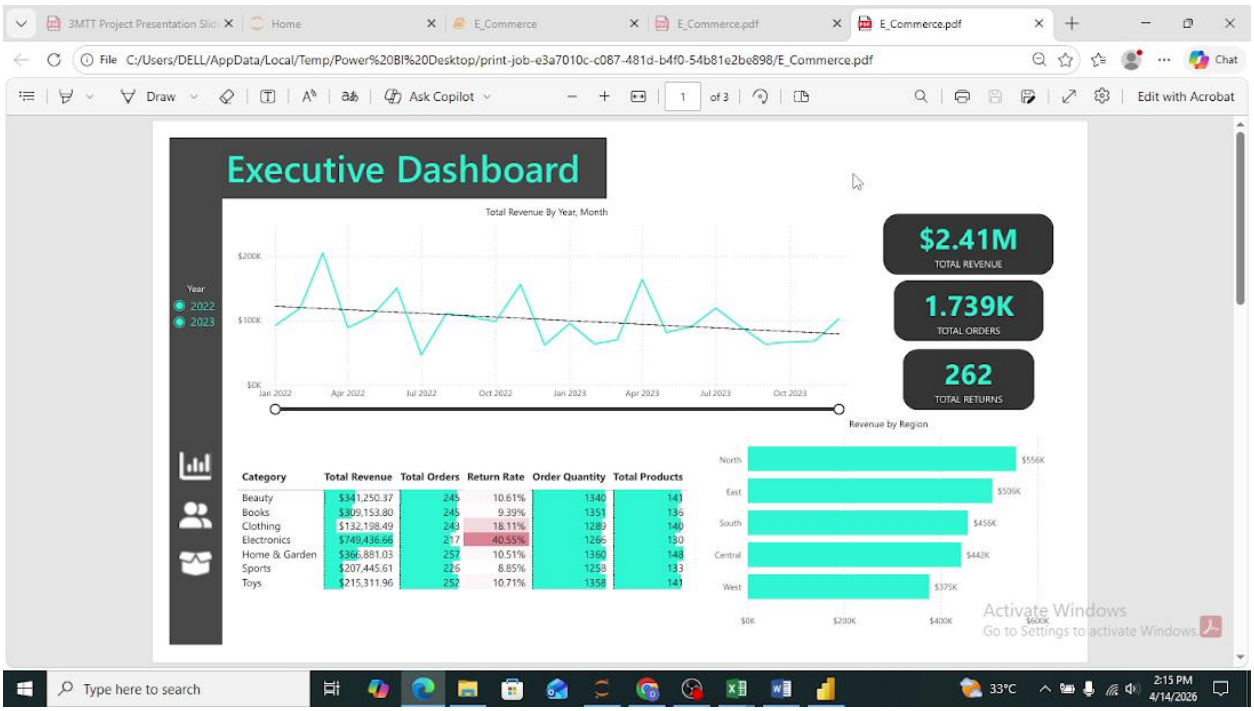
The dataset was analyzed to uncover trends and patterns:

- Revenue distribution by City and Product Category
- Customer behavior based on age groups
- Time-based sales trends
- Impact of discounts on revenue

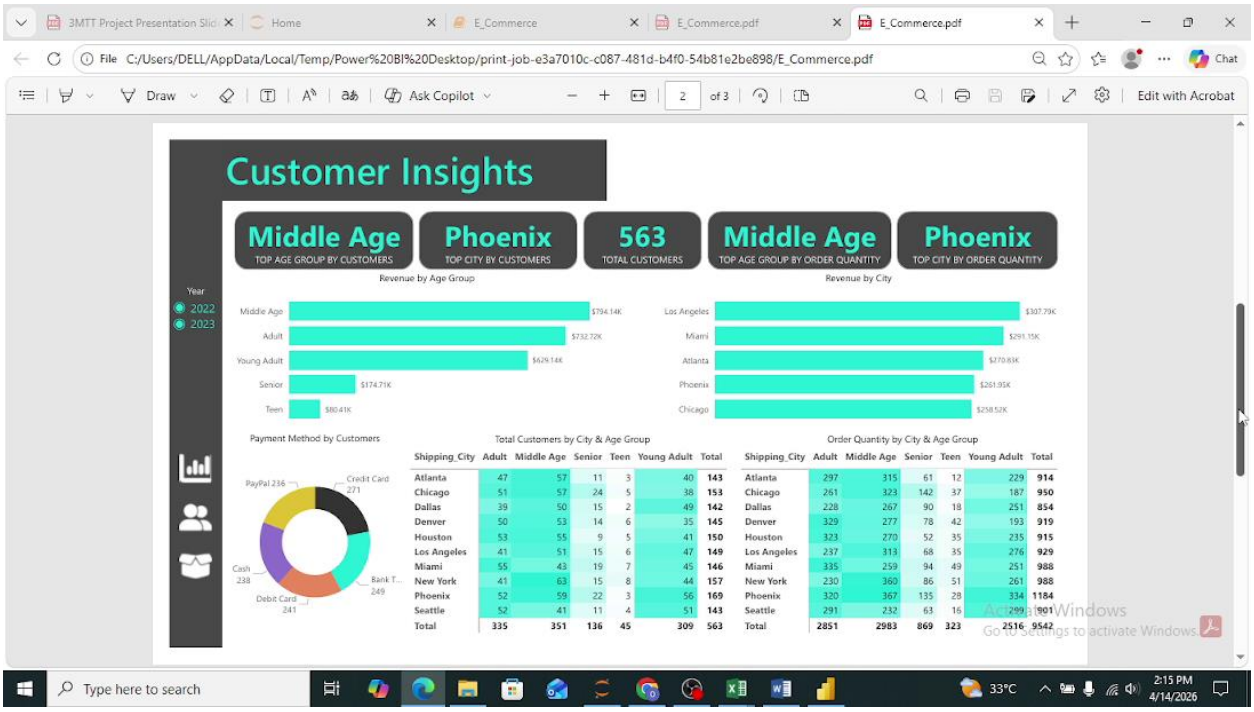
DASHBOARD DEVELOPMENT

An interactive Power BI dashboard was created featuring:

- KPI Cards (Revenue, Orders, Returns)
- Sales Performance Charts
- Customer Insights Visuals
- Time-Based Analysis
- Product Performance Metrics

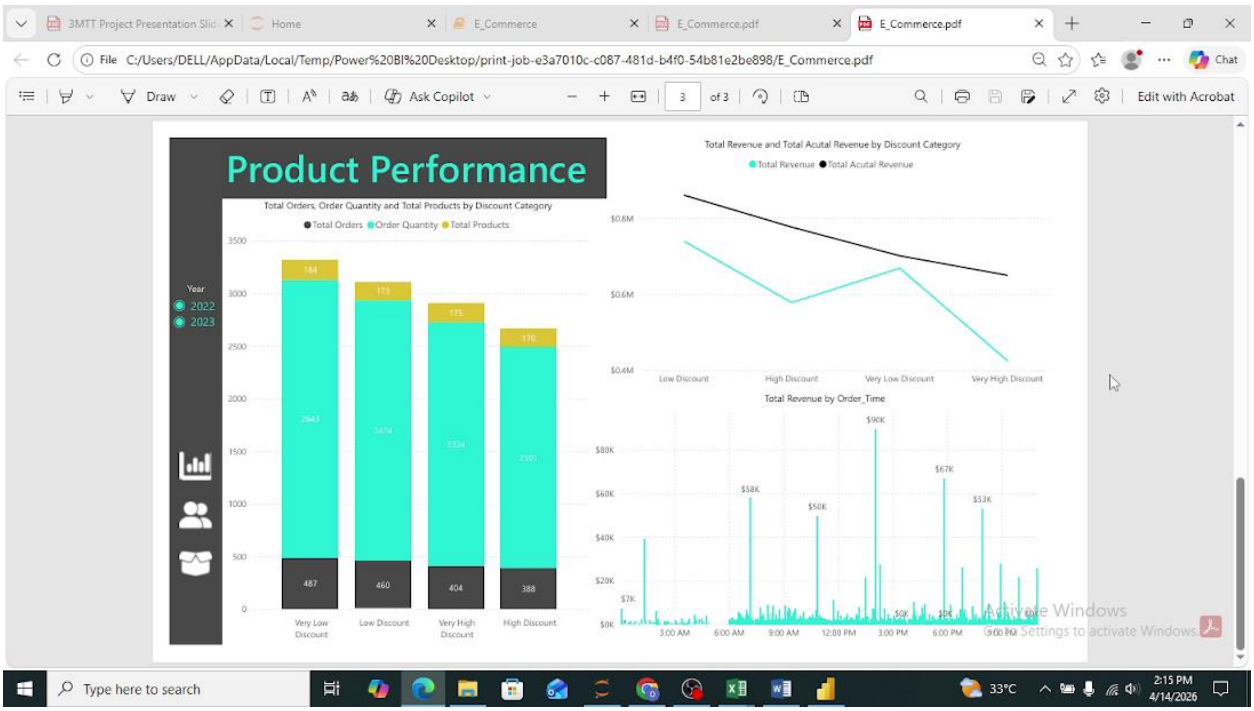


Category	Total Revenue	Total Orders	Return Rate	Order Quantity	Total Products
Beauty	$341,250.37	245	10.61%	1340	141
Books	$309,153.80	245	9.39%	1351	136
Clothing	$132,198.49	249	18.11%	1289	140
Electronics	$749,436.66	217	40.55%	1266	130
Home & Garden	$366,881.03	257	10.51%	1360	148
Sports	$207,445.61	226	8.85%	1259	133
Toys	$215,311.96	252	10.71%	1358	141



Shipping_City	Adult	Middle Age	Senior	Teen	Young Adult	Total
Atlanta	47	57	11	3	40	143
Chicago	51	57	24	5	38	153
Dallas	39	50	15	2	49	142
Denver	50	53	14	6	35	145
Houston	53	55	9	5	41	150
Los Angeles	41	51	15	6	47	149
Miami	55	43	19	7	45	146
New York	41	63	15	8	44	157
Phoenix	52	59	22	3	56	169
Seattle	52	41	11	4	51	143
Total	335	351	136	45	309	563

Shipping_City	Adult	Middle Age	Senior	Teen	Young Adult	Total
Atlanta	297	315	61	12	229	914
Chicago	261	323	142	37	187	950
Dallas	228	267	90	18	251	854
Denver	329	277	78	42	193	919
Houston	323	270	52	35	235	915
Los Angeles	237	313	68	35	276	929
Miami	335	259	94	49	251	988
New York	230	360	86	51	261	988
Phoenix	320	367	135	28	334	1184
Seattle	291	232	63	16	299	901
Total	2851	2983	869	323	2516	9542



KEY INSIGHTS

- The regions had relatively equal contributions to the overall revenue
- The Shipping_cities generated relatively equal revenue
- Electronic product category drove the highest sales revenue
- Peak sales occurred around 2:04:31 PM
- The Middle Age, Young Adult and Adult Age_Groups contributed significantly to the overall revenue
- Products with Low Discounts drove the highest revenue
- Electronic products category had the highest return rates

BUSINESS RECOMMENDATIONS

- Optimise marketing efforts across the regions
- Assign Very Low discounts for products to increase revenue
- Improve quality control for high-return products such electronics and clothing
- Target high-value customer age groups such as middle age, young adult and adults

CHALLENGES FACED

- Mixed date formats across the dataset
- Incorrect and unrealistic age values
- Duplicate records affecting data accuracy

FUTURE IMPROVEMENTS

- Implement predictive analytics (sales forecasting)
- Integrate real-time data pipelines
- Enhance dashboard interactivity

TOOLS & TECHNOLOGIES

- Python (Pandas)
- Power BI
- Excel
- DAX

CONCLUSION

This project demonstrates the importance of data cleaning, analysis, and visualization in transforming raw data into actionable insights. The final dashboard provides a clear and interactive way for businesses to monitor performance and make informed decisions.